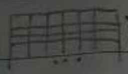
Assignment

data set การสมัครบัตรสินเชื่อจากธนาคารแห่งหนึ่ง

 } 7 ขั้นตอนที่ 1 คำนวณค่าแต่ละ attribute เทียบกับ Class เพื่อหา attribute ที่มีค่า IG มากที่สุด มาเป็น root ของ Decision Tree

Age Credit_card : yes $\frac{3}{7} \approx 0.43$ Credit_card : no $\frac{4}{7} = 0.57$

$20..40 = \frac{3}{7} \approx 0.43$ $>40 = \frac{2}{7} \approx 0.29$ $<= 20$ $\frac{2}{7} \approx 0.29$

$20..40 - y = \frac{2}{3} \approx 0.67$ $>40 - y = \frac{1}{2} = 0.5$ $<= 20 - y = \frac{0}{2} = 0$

$20..40 - n = \frac{1}{3} \approx 0.33$ $>40 - n = \frac{1}{2} = 0.5$ $<= 20 - n = \frac{2}{2} = 1$

$$\text{entropy}(\text{Parent}) = -P(y) \times \log_2 P(y) + -P(n) \times \log_2 P(n)$$
$$= -[0.43 \times \log_2(0.43) + 0.57 \times \log_2(0.57)]$$
$$= -[0.43 \times (-1.22) + 0.57 \times (-0.81)]$$
$$= 0.99$$

$$\text{entropy}(\text{Age} = 20..40) = -[0.67 \times \log_2(0.67) + 0.33 \times \log_2(0.33)]$$
$$= -[0.67 \times (-0.58) + 0.33 \times (-1.60)]$$
$$= 0.91$$

$$\text{entropy}(\text{Age} > 40) = -[0.5 \times \log_2(0.5) + 0.5 \times \log_2(0.5)]$$
$$= 1$$

$$\text{entropy}(\text{Age} <= 20) = -[0 \times \log_2(0) + 1 \times \log_2(1)]$$
$$= -[0 \cdot 1 + 1 \cdot 0] = 0$$

$$IG(\text{Parent}, \text{child}) = \text{entropy}(\text{Parent}) - [P(20..40) \times \text{entropy}(20..40) + P(>40) \times \text{entropy}(>40)$$
$$+ P(<=20) \times \text{entropy}(<=20)]$$
$$= 0.99 - [(0.43 \times 0.91) + (0.29 \times 1) + (0.29 \times 0)]$$
$$= 0.99 - 0.68 = 0.31$$

Income Credit_card : yes $\frac{3}{7} \approx 0.43$ Credit_card : no $\frac{4}{7} = 0.57$

High $= \frac{3}{7} \approx 0.43$ Medium $= \frac{2}{7} \approx 0.29$ Low $= \frac{2}{7} \approx 0.29$

$H - y = \frac{2}{3} \approx 0.67$ $M - y = \frac{1}{2} = 0.5$ $L - y = \frac{0}{2} = 0$

$H - n = \frac{1}{3} \approx 0.33$ $M - n = \frac{1}{2} = 0.5$ $L - n = \frac{2}{2} = 1$

Mon Tue Wed Thu Fri Sat Sun

$\text{entropy}(\text{Income} = H) = -[0.67 \times \log_2(0.67) + 0.33 \times \log_2(0.33)]$

$= 0.91$

$\text{entropy}(\text{Income} = M) = -[0.5 \times \log_2(0.5) + 0.5 \times \log_2(0.5)]$

$= 1$

$\text{entropy}(\text{Income} = L) = -[0 \times \log_2(0) + 1 \times \log_2(1)]$

$= 0$

$IG(\text{Parent}, \text{child}) = 0.99 - [(0.43 \times 0.91) + (0.29 \times 1) + (0.29 \times 0)]$

$= 0.31$ #

Status Credit_card = yes Credit_card = no

Single = $4/7$ = 0.57 marry = $3/7$ = 0.43

S - y = $2/4$ = 0.5 M - y = $1/3$ = 0.33

S - n = $2/4$ = 0.5 M - n $2/3$ = 0.67

$\text{entropy}(\text{Status} = S) = -[0.5 \times \log_2(0.5) + 0.5 \times \log_2(0.5)]$

$= 1$

$\text{entropy}(\text{Status} = M) = -[0.33 \times \log_2(0.33) + 0.67 \times \log_2(0.67)]$

$= 0.91$

$IG(\text{Parent}, \text{child}) = 0.99 - [(0.57 \times 1) + (0.43 \times 0.91)]$

$= 0.03$ #

Education Credit_card = yes Credit_card

Master = $3/7$ = 0.43 Bachelor = $4/7$ = 0.57

M - y = $3/3$ = 1 B - y = $0/4$ = 0

M - n = $0/3$ = 0 B - n = $4/4$ = 1

$\text{entropy}(\text{Education} = M) = -[1 \times \log_2(1) + 0 \times \log_2(0)]$

$= 0$

$\text{entropy}(\text{Education} = B) = -[0 \times \log_2(0) + 1 \times \log_2(1)]$

$= 0$

$IG(\text{Parent}, \text{Child}) = 0.99 - [(0.43 \times 0) + (0.57 \times 0)]$

$= 0.99$ #

Summarize Results

IG - Age = 0.31

IG - Income = 0.31

IG - Status = 0.03

IG - Education = 0.99 \\ Root node

Dicision Rules

Education

Master — Bachelor

Master → Yes

Bachelor → No